

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Machine Learning [CSE-4032]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A weather forecasting system aims to predict the daily weather condition (sunny, cloudy, or rainy) based on observed weather features like temperature, humidity, and wind speed. How can a Hidden Markov Model (HMM) be applied to model this system and predict the most likely sequence of weather conditions? Discuss the role of hidden states, observed emissions, transition probabilities, and emission probabilities in this context. (4)
 - A)
 - B)
 - C)
- 2) A machine learning model is trained to classify emails as spam or not spam. Out of 10,000 emails, 8,000 are actually not spam and 2,000 are actually spam. The model correctly identifies 7,200 non-spam emails as non-spam and 1,600 spam emails as spam. (5)
 - A)
 - B)

A hospital wants to classify patients as either having a heart condition (class 1) or not having a heart condition (class 0) based on two features: Age and Cholesterol Level. The hospital decides to use the K-Nearest Neighbors (KNN) algorithm with $k=3$. The following table represents the data of 6 existing patients, where each row shows the age, cholesterol level, and whether the patient has a heart condition:

Patient	Age	Cholesterol Level	Heart Condition(0 = No, 1 = Yes)
A	45	200	1
B	50	220	1
C	40	180	0
D	35	170	0
E	60	240	1
F	55	230	1

A new patient arrives with an Age of 42 and Cholesterol Level of 190. Using Euclidean distance, classify the new patient using the KNN algorithm with $k=3$. Show your calculations for the distances and the final classification result.

- 3)
 - A) A telecommunications company wants to predict which customers are likely to churn (cancel their service). They have a dataset of customer information, including demographic data, usage patterns, and billing history. To build a predictive model, they can use logistic regression. Explain how applying regularization (L1 or L2) would improve model performance. Discuss the trade-offs between using L1 and L2 regularization for this case. (5)
 - B) A data scientist is tasked with building a model to predict house prices. They have a dataset containing features like square footage, number of bedrooms, and location. How can the data scientist balance the trade-off between underfitting and overfitting when developing this model? Illustrate specific techniques that can be used to optimize the model's performance. (5)
- 4)
 - A) Explain the concept of Generative Adversarial Networks (GANs). Describe the roles of the generator and discriminator in the GAN architecture. How do these two components interact during the training process? Provide a real-world example to illustrate the application of GANs. (5)
 - B) Consider a dataset of 10 data points in a 2D space.
 $X = [[4, 2], [3, 2], [2, 3], [8, 7], [8, 8], [7, 8], [7, 7], [6, 7], [6, 6], [5, 6]]$.
 use the DBSCAN clustering algorithm to identify clusters within this dataset with $\text{minpts}=3$ and $\epsilon=1.9$. (5)
- 5) Analyze the limitations of traditional tabular methods for value function estimation in reinforcement learning, particularly in the context of large or continuous state spaces. (5)

- A) Discuss various approaches to address these limitations. Discuss the potential benefits and challenges of such approaches, providing specific examples to illustrate your points.
- B) Explain the concept of Principal Component Analysis (PCA) in the context of dimensionality reduction. How does PCA work, and what are the key steps involved in its implementation? (5)

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